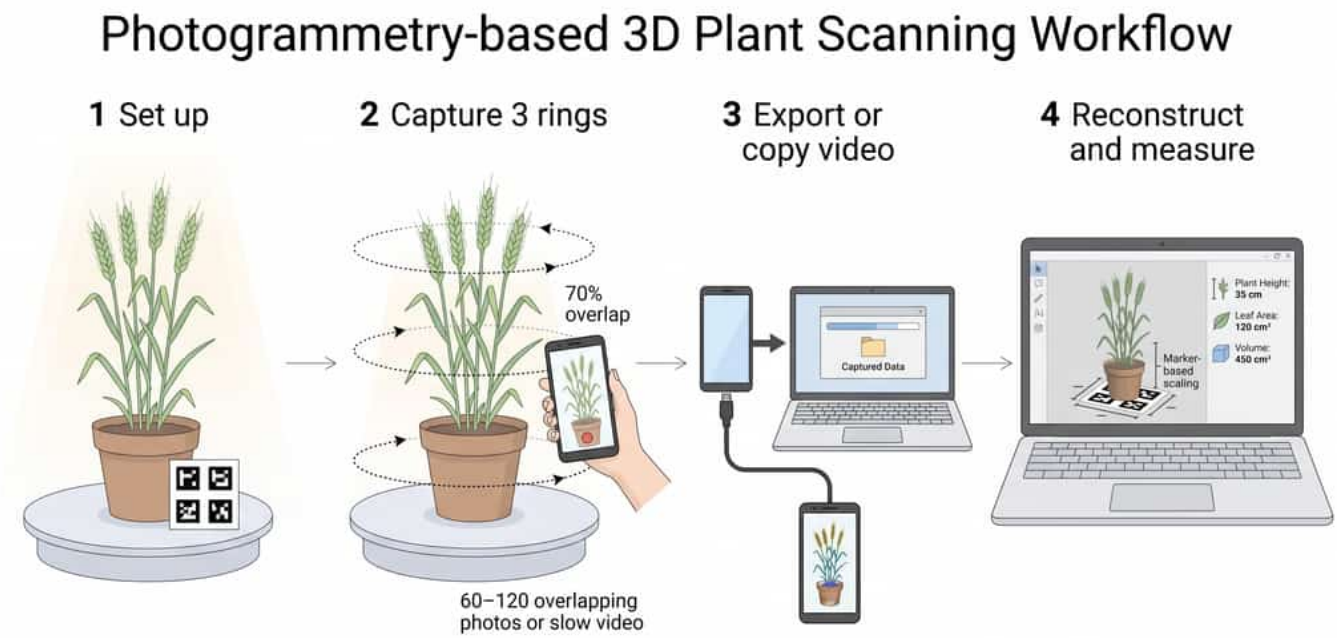
SOP-2 Android phone

Photogrammetry from photos or video, or an app export, scaled with ArUco markers

SOP-2 · PhenoForge capture SOP · version 1.0 (October 2026)



What you need

Checklist

- Any recent Android phone with a good main camera (4K video)
- Option A: a scanning app that exports OBJ / GLB / PLY or Gaussian-splat PLY (e.g. Polycam, Scaniverse, KIRI Engine, RealityScan)
- Option B: just the camera app (photos or video) – PhenoForge reconstructs with COLMAP
- ArUco card (50 mm markers), turntable optional
- USB cable to copy files

Capture

1 Lock the camera
Turn off HDR, beauty and AI scene modes. Use the main (1×) lens only. Tap and hold on the plant to lock focus and exposure (AE/AF lock).

2 Capture three rings
Photos: 20–40 per ring with ~70 % overlap, at low, middle and top-down heights (60–120 in total). Video: 4K 30 fps, one slow circle per ring, ~40 s per circle, no fast turns.

3 Keep the markers in view
The ArUco card must appear in most photos, flat and sharp.

4 Export or copy
Option A: export the app's model as OBJ/GLB/PLY. Option B: copy the photo folder or the video to the PC by USB (DCIM/Camera).

In PhenoForge

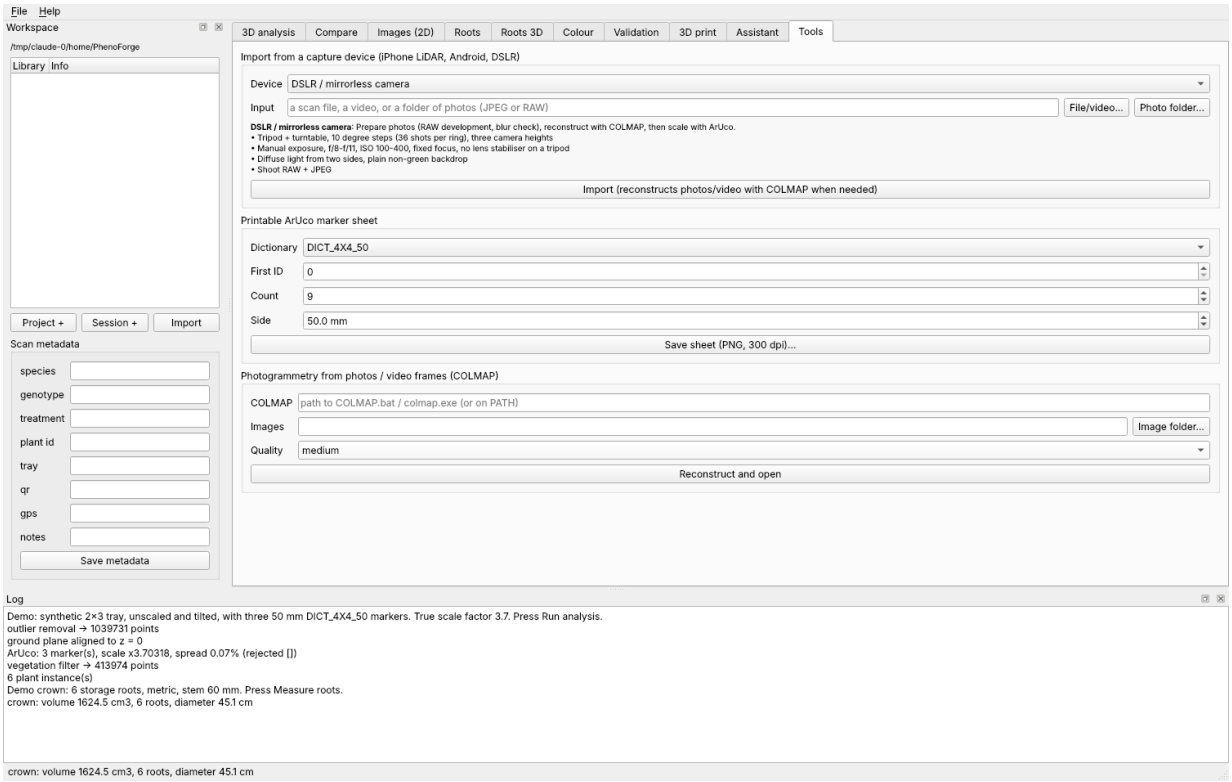
1 **Import**
Tools > Import from a capture device, device *Android phone*. Choose the app export, the video (**File/video…**) or the photo folder (**Photo folder…**). For video the sharpest ~120 frames are picked automatically; blurred or over-exposed photos are rejected and listed in the log.

2 **Reconstruct**
Photos and video are reconstructed with COLMAP (set its path once in Tools; install from colmap.github.io). Use quality *medium* first; *high* needs an NVIDIA GPU and more time.

3 **Scale with ArUco**
The model opens in the 3D analysis tab with scale source **ArUco markers** preselected. Type the measured marker size.

4 **Segment**
Section 3: *single* for one plant, *watershed* for a tray of touching plants (enter the number of plants), or *grid* with rows, columns and pot pitch for a bench of equally spaced pots.

5 **Run and export**
Press **Run analysis**. Results appear in the Scan / Per plant / Leaves tables; **Save results** writes JSON and CSV. Colour the view by Height, Greenness or Plants to check the segmentation.



Tools tab: device import with capture tips for the chosen device.

Troubleshooting

Symptom	Likely cause	Fix
COLMAP registers few images	Too little overlap or motion blur	Move slower; more photos per ring; better light
Plant is doubled or smeared	Plant moved (wind, touching)	Capture indoors; do not touch the pot
No markers detected	Card curved, too small or out of view	Flat card, 50 mm markers, in view from all sides
Scale differs between markers	Marker not flat or partly hidden	Rejected automatically above 4 %; reposition the card